

Music Graph Analysis

E1. Dataset and feature extraction

The dataset consists of 2,000 tracks by 673 artists from the MTG-Jamendo mood/theme subset [1, 2]. The aim is to study acoustic similarity, connectivity and spectral groups, and relate those groups to music tags. Using seed 521, 12 of 100 shuffled feature archives were selected, and 2,000 eligible tracks were sampled from those archives. This is a two-stage pilot sample, not a uniform sample of the entire collection.

Each track is represented by 100 features from the supplied Essentia/AcousticBrainz statistics:

Group	Features	Dimension
Timbre	13 MFCC means and 13 standard deviations	26
Pitch class	36 HPCP means and 36 standard deviations	72
Rhythm	BPM and onset rate	2

MFCC denotes mel-frequency cepstral coefficients; HPCP denotes harmonic pitch-class profile. Standard deviations are the square roots of the MFCC covariance diagonal and HPCP variances.

All features are standardized using population standard deviations. A group of size m_g is scaled by $1/\sqrt{m_g}$, giving each group equal total variance before PCA. With μ_j and s_j denoting feature means and standard deviations,

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{s_j \sqrt{m_{g(j)}}}, \quad f_i = (\tilde{x}_i - \bar{\tilde{x}})V_{32}.$$

Full-SVD PCA retains 32 components without whitening, preserving 94.42% of the group-weighted variance. The retained dimension is a hyperparameter. Genre and other tags are excluded from the feature vectors.

E2. Pairwise interaction matrix

For the PCA features $f_i \in \mathbb{R}^{32}$, define Euclidean distances and Gaussian similarities by

$$R_{ij} = \|f_i - f_j\|_2, \quad W_{ij} = \exp\left(-\frac{R_{ij}^2}{2\sigma^2}\right) \quad (i \neq j), \quad W_{ii} = 0.$$

The bandwidth $\sigma = 1.24392285$ is the median tenth-nearest-neighbor distance, held fixed for all graphs. The matrices are symmetric and 2000×2000 , covering 1,999,000 unordered pairs. The median distance is 2.253910 and the median off-diagonal similarity is 0.193678. Larger weights indicate greater acoustic similarity. Three distinct track pairs have identical feature vectors and similarity one; their track identifiers remain separate.

E3. kNN graphs and connected components

For every integer $k = 2, \dots, 64$, let $N_k(i)$ contain the k nearest non-self neighbors of vertex i . Distance ties are resolved by node index. The weighted, undirected union-kNN adjacency is

$$A_{ij}^{(k)} = W_{ij} \mathbf{1}\{j \in N_k(i) \text{ or } i \in N_k(j)\}, \quad A_{ii}^{(k)} = 0.$$

Thus an edge is kept if either endpoint selects the other; a vertex's degree may exceed k . Because the Gaussian kernel preserves distance ordering, this is equivalent to retaining the largest non-self similarities before symmetrization.

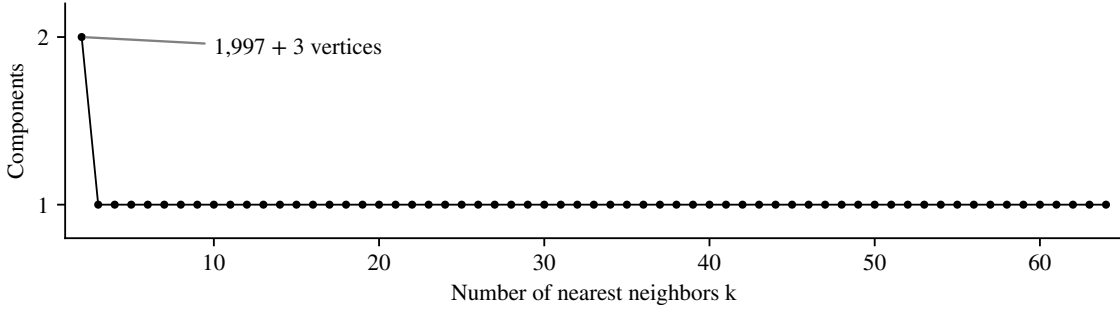


Figure 1: Connected-component count for all $k = 2, \dots, 64$.

At $k = 2$, the graph has 3,311 edges and components of sizes 1,997 and 3. All graphs with $k \geq 3$ are connected, so the minimum connectivity threshold is $k_c = 3$. Fixed neighbor ordering gives nested edge sets as k increases. The 1,997-vertex component, containing 99.85% of the sample, is the large component analyzed in E4 and E5.

E4. Degree and local clustering distributions

Combinatorial statistics use the binary topology $B_{ij} = \mathbf{1}\{A_{ij} > 0\}$. For degree d_i and the number T_i of triangles incident on vertex i ,

$$d_i = \sum_j B_{ij}, \quad C_i = \frac{2T_i}{d_i(d_i - 1)}.$$

Every vertex has degree at least two, so all denominators are nonzero. Figure 2 shows the empirical probability masses.

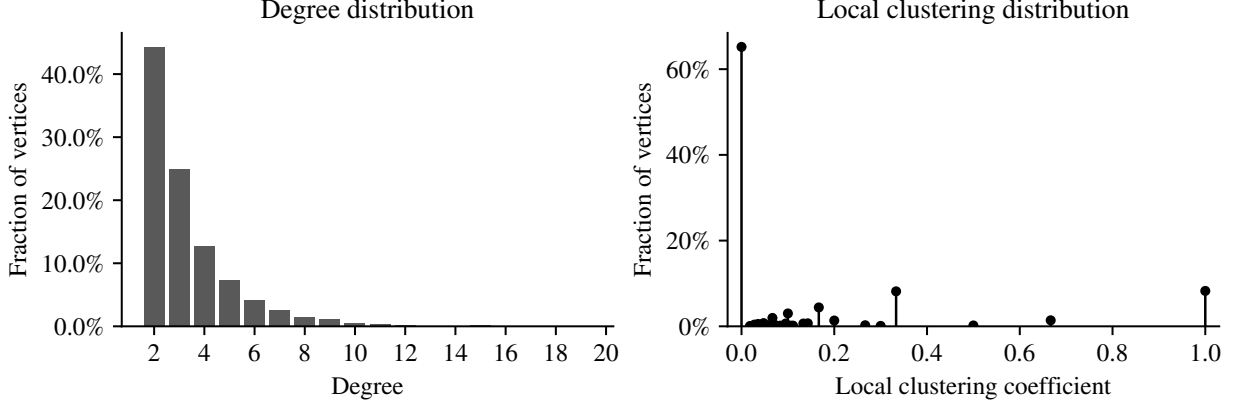


Figure 2: Degree and local clustering distributions of the large $k = 2$ component. Clustering values are grouped by their exact rational values.

Statistic	Value	Statistic	Value
Vertices / edges	1,997 / 3,308	Mean C_i	0.140104
Degree range	2–19	Median C_i	0
Mean / median degree	3.312969 / 3	Fraction with $C_i = 0$	65.20%
Degree mode	2	Fraction with $C_i = 1$	8.26%

The degree distribution is discrete and right-skewed, with a dominant mode at 2: 44.32% of vertices have degree 2 and 81.97% have degree at most 4. The observed tail alone does not establish a power law or other parametric family. The clustering distribution has a large point mass at zero and several rational-valued spikes. Such spikes are natural at low degrees: when $d_i = 2$, only $C_i = 0$ and $C_i = 1$ are possible. The three-vertex component is a triangle, whose degree and clustering distributions are point masses at 2 and 1, respectively.

E5. Laplacian spectral analysis

Normalized Laplacian and the Fiedler value

Spectral analysis uses the original Gaussian weights. Let $D_{ii} = \sum_j A_{ij}$ be weighted degree (strength). The symmetric normalized Laplacian is

$$\hat{L} = I - D^{-1/2} A D^{-1/2}.$$

Within a connected component, $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_n \leq 2$, with zero eigenvector proportional to $(\sqrt{D_{11}}, \dots, \sqrt{D_{nn}})$.

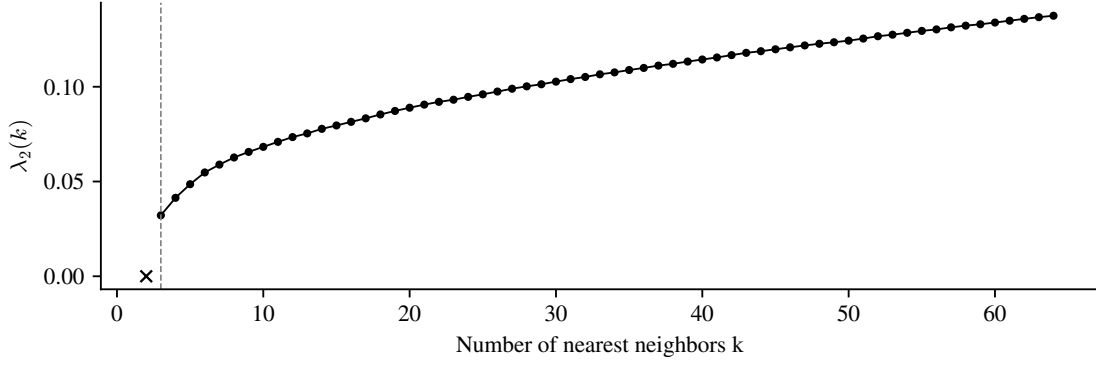


Figure 3: Full-graph Fiedler value. The graph becomes connected at $k_c = 3$.

For the full 2,000-vertex graph, $\lambda_2(k)$ increases at every step from 0.03215373 at $k = 3$ to 0.13746997 at $k = 64$, indicating stronger normalized spectral connectivity. This is an empirical observation, not a general monotonicity guarantee: adding edges changes both A and D . At $k = 2$, the full graph has $\lambda_2 = 0$, whereas its large connected component has $\lambda_2 = 0.01816991$.

Low spectrum and a separated simple eigenvalue

Figure 4 shows the 32 smallest eigenvalues of the large component. To select a separated nonzero eigenvalue, maximize

$$g_j = \min\{\lambda_j - \lambda_{j-1}, \lambda_{j+1} - \lambda_j\}, \quad j = 2, \dots, 32.$$

The 33rd eigenvalue supplies the last candidate's right-hand gap.

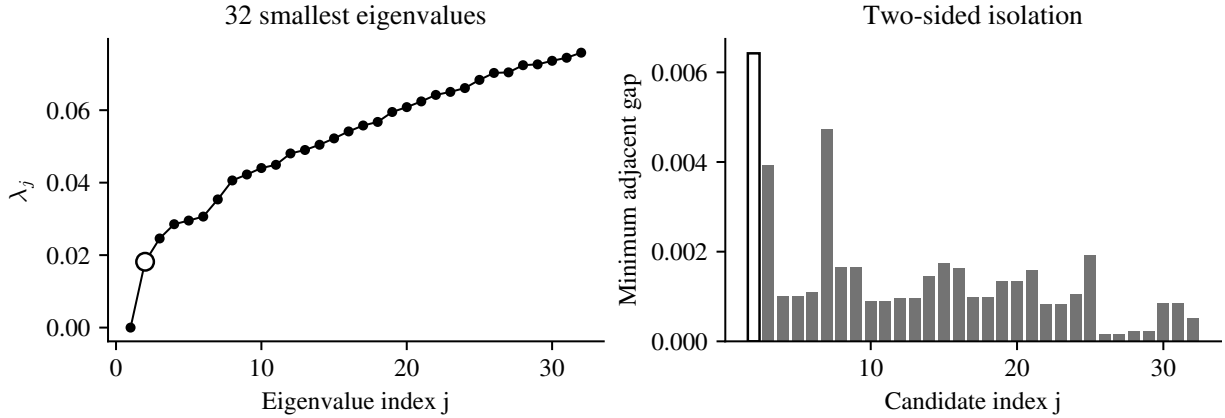


Figure 4: The 32 lowest eigenvalues and two-sided isolation gaps.

The selected eigenvalue is $\lambda_2 = 0.0181699110$. Its left and right gaps are 0.0181699110 and 0.0064237727, respectively; the smaller gap is 35.35% of λ_2 . The largest eigenpair residual is 1.12×10^{-15} , far below these gaps, supporting numerical simplicity and separation. The small weighted triangle has eigenvalues 0, 1.47716388, and 1.52283612.

Three-dimensional embedding and nodal domains

Use the coordinates $(q_2(i), q_3(i), q_4(i))$ from orthonormal eigenvectors. Their eigenvalues are 0.01816991, 0.02459368, and 0.02853374, respectively. Writing $Q = [q_2 \ q_3 \ q_4]$ gives

$$\hat{L}Q = Q \operatorname{diag}(\lambda_2, \lambda_3, \lambda_4),$$

so these vectors span a three-dimensional invariant subspace. A strong nodal domain is a connected component of the subgraph induced by $q_2 > 0$ or $q_2 < 0$. Both sign-induced subgraphs are connected, giving domains of 963 and 1,034 vertices. No eigenvector entries are numerically zero.

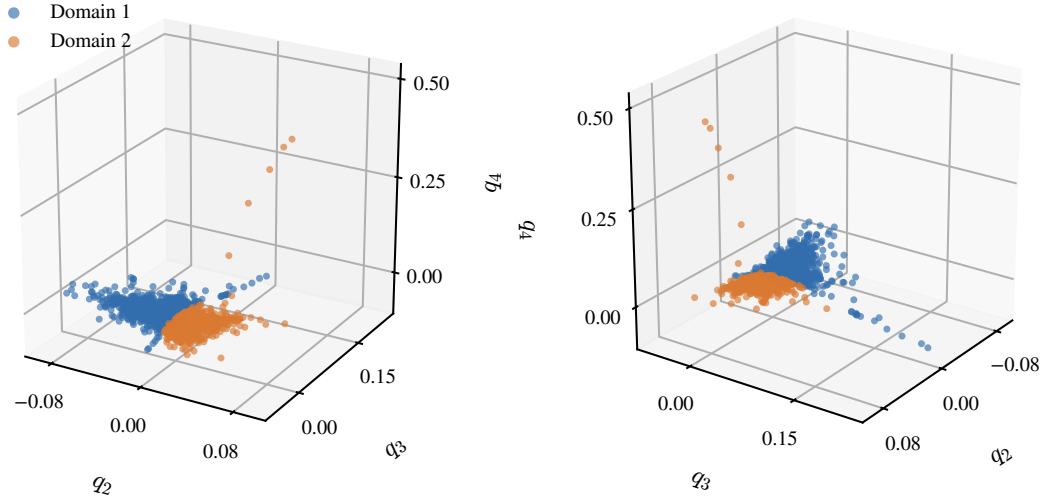


Figure 5: Two views of the embedding in (q_2, q_3, q_4) , colored by the nodal domains of q_2 : domain 1 (blue, 963 vertices) and domain 2 (orange, 1,034 vertices).

The domains are joined by 175 edges, with normalized cut 0.104903. They are connected parts of the large component, not separate components of the original graph. Reversing the eigenvector sign exchanges the labels but preserves membership.

Relationship to the original data

The strongest raw-feature association is onset rate. Domain means account for 44.97% of its total variance, measured by

$$\eta^2 = \frac{\sum_g n_g (\bar{x}_g - \bar{x})^2}{\sum_i (x_i - \bar{x})^2}.$$

Measure	Domain 1	Domain 2
Mean onset rate (events/s)	2.433	4.385
Mean BPM	128.62	116.87
MFCC mean 01 (standardized mean)	0.49	-0.46

Domain 2 has denser onsets but lower average BPM, so the partition is not simply fast versus slow music. Several MFCC means also differ; for example, MFCC mean 01 has $\eta^2 = 0.2264$, indicating a timbral contribution.

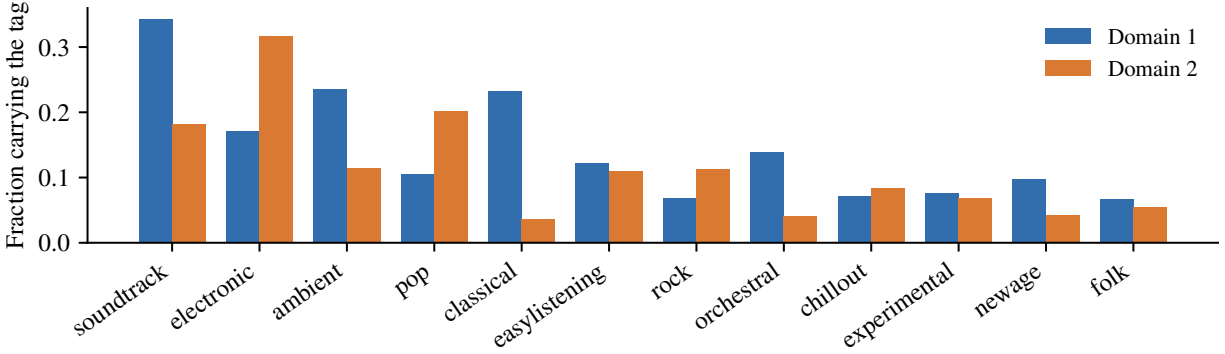


Figure 6: Prevalence of the 12 most common genre tags within each nodal domain. Tracks may carry multiple labels.

Domain 1 has more classical, soundtrack, ambient and orchestral labels; domain 2 has more electronic and pop labels. Classical prevalence is 23.16% versus 3.58%, while electronic prevalence is 17.13% versus 31.72%. The domains remain musically mixed. Acoustic associations are descriptive because those features also define the graph; tags provide an interpretation, not an independent classification test.

E7. Discussion

The graph is almost connected at $k = 2$ and fully connected at $k = 3$, despite low degrees and little triangle closure. Its spectral partition captures differences in onset density, timbre and genre composition. Increasing k strengthens normalized spectral connectivity, but does not necessarily improve musical interpretation.

The two-stage sample, repeated artists and representation choices limit generalization. High retained PCA variance does not guarantee preservation of every nearest neighbor. Artist-aware resampling and parameter sensitivity analysis would help assess robustness. These findings describe the specified graph, rather than pure genre classes or predictive performance.

Reproducibility. Scripts and LaTeX source are in `code/`; numerical outputs are in `results/`. Rebuild the report with `python code/build_report.py`.

References

- [1] D. Bogdanov, M. Won, P. Tovstogan, A. Porter, and X. Serra. *The MTG-Jamendo Dataset for Automatic Music Tagging*. Machine Learning for Music Discovery Workshop, ICML, 2019. <https://hdl.handle.net/10230/42015>.
- [2] Music Technology Group. MTG-Jamendo dataset: metadata, features and documentation. <https://github.com/MTG/mtg-jamendo-dataset>.